ChE 313 – Transport phenomena III –Spring 2025

Student Learning Outcome 1

Rubric for assessment problem on the first exam (Problem #3-(2) & (3))

Derive the equation to calculate the steady-state mass flux through a membrane.

SLO 1 Performance Indicator (PI)	Level 3: Satisfactory	Level 2: Developing	Level 1: Unsatisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem D(a), 10 pts	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (4-5 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (2-3 pts)	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (0-2 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem D(b), 10 pts	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (11-15 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (5-10 pts)	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (0-4 pts)

Averages:

Performance Indicators: (1) 2.40 (2) 2.35 Overall Average: 2.38

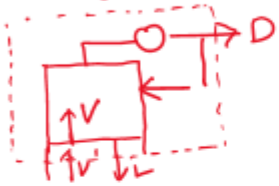
Average scores indicate moderate overall performance with noticeable variability among students. While a portion of the class demonstrated satisfactory ability in formulating governing equations and applying transport principles, a significant subset of students remained at the developing level, particularly in translating physical systems into mathematically consistent mass balance expressions. Students generally performed better on the application of mathematical tools (algebra, calculus, and differential equations) once the governing equations were correctly established. However, difficulties persisted in the initial problem formulation stage.

To improve student outcomes, emphasis will be placed on structured problem formulation, including stepwise development of mass balance equations.

2. (80 pts) We are separating acetone and ethanol in a distillation column with a total condenser. Instead of using a reboiler, saturated pure steam at 1.0 atm is sparged directly into the bottom of the column to provide boilup (called *direct* or *open steam*). The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid that is 60.0 mol% acetone and has a feed rate of F1 = 200.0 kmol/hr. The second feed is 40.0 mol% acetone, has a feed rate of F2 = 300.0 kmol/hr, and is a two-phase mixture that is 20% vapor. Assume CMO is valid. Bottoms product is 2.0 mol% acetone, and distillate product is 80.0 mol% acetone. External reflux ratio L/D = 1.5. The ratio bottoms flow rate/steam flow rate = B/S = 5.0.

3) (5 pts) Draw four mass balance envelopes for each section. Draw separately from (1). Then, write down mass balance for each section.

Enriching section



$V' = L + D$

Feed 1



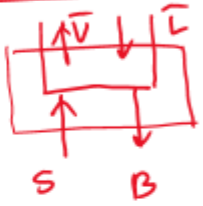
$F_1 + V' + L = V + L'$

Feed 2



$F_2 + \bar{V} + L' = V' + \bar{L}$

Bottom Section



$\bar{V} + B = \bar{L} + S$

4) (15pts) Find D, B, S, L, V, V', L', $\bar{L}$, and $\bar{V}$.

- $F_1 + F_2 + 0.2B = D + B$, $F_1 + F_2 = D + 0.8B$ ($B/S = 5.0$)

500

- $F_1 z_1 + F_2 z_2 = D x_D + B x_B$, $240 = 0.8D + 0.02B$

$240 = (0.8)\cdot(500 - 0.8B) + 0.02B$, $240 = 400 - 0.64B + 0.02B$

$B = 258.07$ kmol/hr, $D = 293.54$ kmol/hr

- $L = \left(\frac{L}{D}\right) D = 1.5 \times 293.54 = 440.32$ kmol/hr

$S = 258.07/5.0 = 51.62$ kmol/hr

$V = L + D = 440.32 + 293.54 = 733.86$ kmol/hr

$V' = V = 733.86$ kmol/hr

$L' = L + F_1 = 440.32 + 200 = 640.32$ kmol/hr

$\bar{L} = L' + 0.8F_2 = 640.32 + 0.8 \times 300 = 880.32$ kmol/hr

$V' = \bar{V} + 0.2F_2 \Rightarrow \bar{V} = V' - 60 = 733.86 - 60 = 673.86$ kmol/hr